

Square Roots

The square root is the reverse of squaring. Learn the symbol, the prime-factorisation method, how to estimate roots of non-perfect squares, and the "smallest number to multiply/divide" problems that show up for 3 marks.

What is a square root?

The idea: Squaring a number and taking its square root are **opposite** operations. If $y = x^2$, then x is the **square root of y** . The symbol is $\sqrt{}$.

$$7^2 = 49 \iff \sqrt{49} = 7 \quad (\text{a square of area 49 has side 7})$$

Positive and negative roots

Because $8 \times 8 = 64$ and $(-8) \times (-8) = 64$, every perfect square has **two** square roots — one positive, one negative.

$$\sqrt{64} = \pm 8 \quad | \quad \text{in this chapter we use only the **positive** root, so we write } \sqrt{64} = 8$$

Method 1 — Square root by prime factorisation

Break the number into prime factors, **pair** identical primes, and take **one factor from each pair**. Their product is the square root.

Quick test for a perfect square: if every prime factor can be grouped into **pairs** with none left over, the number is a perfect square.

Method 2 — Repeated subtraction (small numbers)

Subtract 1, 3, 5, 7, ... until you reach 0. The count of subtractions is the square root (covered in Concept 1).

Estimating the root of a non-perfect square

Trap it between the two nearest perfect squares you know, then narrow down. (Useful when the number is not a perfect square.)

Smallest number to multiply / divide: If one prime is left unpaired, multiplying or dividing by that prime fixes the pairing and makes a perfect square — a common 3-mark question.

Prerequisite refresher: *Prime factorisation* = writing a number as a product of primes (e.g. $36 = 2 \times 2 \times 3 \times 3$). A *prime* has exactly two factors: 1 and itself.

Where marks slip away

Trap 1 — Taking the whole pair. From a pair (3×3) you take **one** 3 for the root, not 3×3 . $\sqrt{(3 \times 3)} = 3$.

Trap 2 — Forgetting a leftover prime. If a prime can't be paired, the number is **not** a perfect square. Don't force it.

Trap 3 — "Multiply" vs "divide". To remove a single leftover prime you can multiply (to add its pair) or divide (to delete it). Read the question — it usually asks for the **smallest** such number.

Trap 4 — Estimation direction. When estimating, state both neighbours (e.g. $15 < \sqrt{250} < 16$) and pick the closer one — don't just round randomly.

Slip 5 — The $\pm$ sign. Mention that perfect squares have $\pm$ roots, but use the positive root unless asked otherwise.

Model answers — write them like this

Example A. Find $\sqrt{11025}$ by prime factorisation.

Step 1: Prime factorise: $11025 = 3 \times 3 \times 5 \times 5 \times 7 \times 7$.

Step 2: Make pairs: $(3 \times 3) \times (5 \times 5) \times (7 \times 7)$.

Step 3: Take one from each pair: $3 \times 5 \times 7 = 105$.

$$\therefore \sqrt{11025} = 105$$

Example B. Find $\sqrt{4761}$ by prime factorisation.

Step 1: $4761 = 3 \times 3 \times 23 \times 23$.

Step 2: Pairs: $(3 \times 3) \times (23 \times 23)$.

Step 3: One from each pair: $3 \times 23 = 69$.

$$\therefore \sqrt{4761} = 69$$

**Example C. By what smallest number should 2800 be multiplied to make it a perfect square?
Find the new square root.**

Step 1: $2800 = 2 \times 2 \times 2 \times 2 \times 5 \times 5 \times 7$.

Step 2: Pairs: $(2 \times 2)(2 \times 2)(5 \times 5)$ and a leftover 7 with no partner.

Step 3: Multiply by 7 to give 7 a pair: $2800 \times 7 = 19600$.

Step 4: $\sqrt{19600} = 2 \times 2 \times 5 \times 7 = 140$.

$\therefore$ Multiply by 7; $\sqrt{19600} = 140$

**Example D. By what smallest number should 216 be divided so the quotient is a perfect square?
Find that square root.**

Step 1: $216 = 2 \times 2 \times 2 \times 3 \times 3 \times 3$.

Step 2: Pairs: (2×2) and (3×3) ; leftover 2 and 3 unpaired.

Step 3: Divide by the leftovers $2 \times 3 = 6$: $216 \div 6 = 36$.

Step 4: $\sqrt{36} = 6$.

$\therefore$ Divide by 6; quotient 36; $\sqrt{36} = 6$

Example E. Estimate $\sqrt{250}$.

Step 1: Nearest squares: $15^2 = 225$ and $16^2 = 256$.

Step 2: So $15 < \sqrt{250} < 16$.

Step 3: 256 is much closer to 250 than 225, so $\sqrt{250} \approx 16$ (just under 16).

$\therefore \sqrt{250} \approx 16$

Next: [try the worksheet](#), then [check the answer key](#).